A NEW UPPER BOUND FOR THE DENSITY OF PLANAR SETS AVOIDING UNIT DISTANCES

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Abstract. Let $m_1(\mathbb{R}^2)$ be the supremum of the upper densities of measurable planar sets in which no two points are at distance 1. Ambrus, Csiszárik, Matolcsi, Varga and Zsámboki proved $m_1(\mathbb{R}^2) \le 0.2470$, confirming Erdős's conjecture $m_1(\mathbb{R}^2) < 1/4$, with a linear program whose constraints come from the patterns in which a random copy of a set X_{23} of 23 points meets the set; their computer search for X stopped at 24 points. Adding 7 points to X_{23}, with coordinates in the same field $\mathbb{Q}(\sqrt{3}, \sqrt{11})$, we prove $m_1(\mathbb{R}^2) \le 0.24380$. This improves the published bound; the same group has announced 0.2415 without a published proof. Our proof is a rational certificate for the program, checked in exact and interval arithmetic by two programs written separately. With the transfer theorem of OpenAI, as extended to fractional colourings in an earlier note, the fractional chromatic number of the plane is therefore at least 4.1017. In floating point the value of the program keeps decreasing as points are added, so the method has not reached its limit.

1. The result

A set $A \subseteq \mathbb{R}^2$ *avoids the distance* 1 if $|x - y| \neq 1$ for all $x, y \in A$. Its upper density is $\overline{\delta}(A) = \limsup_{R\to\infty} |A \cap B(0,R)|/|B(0,R)|$, and

$$m_1(\mathbb{R}^2) = \sup\{\overline{\delta}(A) : A \subseteq \mathbb{R}^2 \text{ measurable, avoiding the distance } 1\}.$$

The question was popularized by Leo Moser; Croft's construction gives $m_1(\mathbb{R}^2) \ge 0.22936$ [4], and no better construction is known. Erdős wrote that "it seems very likely that $m_1(\mathbb{R}^2)$ is less than $1/4$" [7]. The upper bounds came from unit-distance graphs with large fractional chromatic number (the Moser spindle gives $2/7$; Bellitto, Pêcher and Sédillot 0.2565 [3]) and from linear programs on the autocorrelation of A: Székely $12/43 \approx 0.2791$ [14], Oliveira Filho and Vallentin 0.2684 [10], Keleti, Matolcsi, Oliveira Filho and Ruzsa 0.2588 [9], Ambrus and Matolcsi 0.2544 [2], and finally Ambrus, Csiszárik, Matolcsi, Varga and Zsámboki, $m_1(\mathbb{R}^2) \le 0.2470$ [1], which settled Erdős's conjecture. In December 2023 Varga announced that the group had improved this to 0.2415 [12]; we found no paper, preprint or certificate for that bound. Their program combines both approaches: for a finite set X, the probabilities of the patterns in which a random congruent copy of X meets A satisfy linear equations ("inclusion–exclusion"), whose coefficients are values of the autocorrelation of A. They found a set X_{23} of 23 points by a beam search over sets of at most 24 points (a week on 128 processors [1, §9]).

Theorem 1. *Every measurable planar set avoiding the distance* 1 *has upper density at most* 0.24380*. That is,* $m_1(\mathbb{R}^2) \le 0.24380$*.*

This is weaker than the announced 0.2415, but, as far as we know, it is the best upper bound with a published proof. The method is entirely that of [1]; what is new is the configuration and the certificate. The proof is the program of [1] for a set X_{30} of 30 points: X_{23} and 7 more points in the same field $\mathbb{Q}(\sqrt{3}, \sqrt{11})$ (Section 3). Section 2 states the certificate that proves the bound

Date: Draft, 11 October 2026.

2020 *Mathematics Subject Classification.* Primary 52C10, 05C15; Secondary 42B10, 90C05.

Key words and phrases. unit distance graph, distance avoiding sets, density, fractional chromatic number, chromatic number of the plane, linear programming bounds, inclusion–exclusion.

These results were obtained with the assistance of an AI system (Claude, by Anthropic), which found the configuration, wrote the programs, ran the checks and helped to draft this note; a second program and a referee report were produced by separate AI agents. The method and the 23-point configuration are those of Ambrus, Csiszárik, Matolcsi, Varga and Zsámboki [1]. No mathematician has checked the note yet. The author is not a professional mathematician and submits it for independent verification.

and Section 4 how it was checked. Section 5 describes how the points were found and what larger sets give in floating point.

Consequences. Write $\chi_{\mathrm{f}}(\mathbb{R}^2)$ for the fractional chromatic number of the unit-distance graph of the plane, that is, the supremum of $\chi_{\mathrm{f}}(G)$ over finite unit-distance graphs G. Averaging translates of a set avoiding 1 over the vertices of G gives $m_1(\mathbb{R}^2) \le 1/\chi_{\mathrm{f}}(G)$; the converse direction is the hard one.

Corollary 2. $\chi_{\mathrm{f}}(\mathbb{R}^2) \ge 1/m_1(\mathbb{R}^2) \ge 4.1017$*, and the measurable fractional chromatic number of the plane is at least* 4.1017*. For every $\varepsilon > 0$ and every large n there are n-point unit-distance graphs whose independence number is below $(0.24380 + \varepsilon)n$.*

Proof. The measurable statement is the averaging argument above in the measurable setting. That $\chi_{\mathrm{f}}(\mathbb{R}^2) = 1/m_1(\mathbb{R}^2)$, and that the least independence ratio of n-point unit-distance graphs tends to $m_1(\mathbb{R}^2)$, is proved in [8, Cors. 2 and 3] from the transfer theorem of OpenAI [11, Thm. 2.3]; insert Theorem 1. □

Before, these bounds were 4.0485 and 0.2470 [1, 8]; Dúcz and Varga had found a finite unit-distance graph with fractional chromatic number above 4.0007 [6]. The same substitution improves the other bounds of [8] that use $m_1(\mathbb{R}^2)$: for instance, for suitable sets D of k integers, $\chi(\mathbb{R}^2, D) \ge m_1(\mathbb{R}^2)^{-k} > 4.1017^k$ [8, Cor. 6].

2. The inclusion–exclusion bound

Let $X = \{x_1, \dots, x_n\} \subseteq \mathbb{R}^2$, $r_{ij} = |x_i - x_j|$, let E_1 be the set of pairs $\{i, j\}$ with $r_{ij} = 1$, and let $\mathcal{S}$ be the family of subsets of $\{1, \dots, n\}$ containing no pair of E_1 (the independent sets of the unit-distance graph of X, including $\emptyset$). Let $\mathcal{C}$ be a list of pairs (I, J) of sequences of distinct indices of the same length with $|x_{I_a} - x_{I_b}| = |x_{J_a} - x_{J_b}|$ for all a, b.

Proposition 3 (Ambrus, Csiszárik, Matolcsi, Varga, Zsámboki [1])**.** *Suppose that real numbers $\psi_0, \psi_T, \lambda_i$ $(1 \le i \le n)$, u_{ij} $(\{i, j\} \notin E_1)$ and w_c $(c \in \mathcal{C})$ satisfy*

$$\Phi(t) = \psi_0 J_0(t) - \sum_i \lambda_i - \sum_{i<j} u_{ij} J_0(r_{ij} t) \;\ge\; 1 \qquad (t \ge 0),$$

$$Q(S) = \psi_T + \sum_{i \in S} \lambda_i + \sum_{i<j,\ i,j \in S} u_{ij} + \sum_{(I,J) \in \mathcal{C}} w_{(I,J)} \big([I \subseteq S] - [J \subseteq S] \big) \;\ge\; 0 \qquad (S \in \mathcal{S}),$$

where the sums of u_{ij} run over the pairs $i < j$ with $\{i, j\} \notin E_1$. Then $m_1(\mathbb{R}^2) \le \psi_T$.

Proof (after [1]*).* It suffices to bound the density δ of a measurable set A avoiding 1 that is periodic for a lattice Λ [1, §2]. Let c_ξ $(\xi \in \Lambda^*)$ be the Fourier coefficients of 1_A on $\mathbb{R}^2/\Lambda$, and κ the image of the measure $\sum_\xi |c_\xi|^2 \delta_\xi$ under $\xi \mapsto 2\pi|\xi|$: a measure on $[0, \infty)$ of total mass δ (Parseval) with $\kappa(\{0\}) = \delta^2$. The radial average of the autocorrelation $\varphi(v) = \delta(A \cap (A - v))$ is $f(r) = \int J_0(rt)\, d\kappa(t)$, and $f(1) = 0$ since A avoids 1. Choose y uniformly in $\mathbb{R}^2/\Lambda$ and $g \in O(2)$ according to Haar measure, and let $S = \{i : y + g x_i \in A\}$. Then $S \in \mathcal{S}$, $\Pr(i \in S) = \delta$, $\Pr(i, j \in S) = f(r_{ij})$, and $\Pr(I \subseteq S) = \Pr(J \subseteq S)$ for $(I, J) \in \mathcal{C}$: some isometry $h = h_0 + c$ maps x_{I_a} to x_{J_a} for every a, and $(y, g) \mapsto (y + gc, g h_0)$ preserves the distribution of (y, g) (reflections are allowed because g ranges over $O(2)$; 9 474 of the 15 709 relations of our certificate for X_{30} need an isometry that reverses orientation). Hence $0 \le \mathbb{E}\, Q(S) = \psi_T + \delta \sum_i \lambda_i + \sum_{i<j} u_{ij} f(r_{ij})$. Integrating $\Phi \ge 1$ against κ, including the atom at 0,

$$\delta = \int d\kappa \le \int \Phi \, d\kappa = \psi_0 f(1) - \delta \sum_i \lambda_i - \sum_{i<j} u_{ij} f(r_{ij}) \le \psi_T. \qquad \square$$

The best such bound for given X and $\mathcal{C}$ is the value of a linear program, whose dual variables are the probabilities of the patterns and the measure κ; for X_{23} it is $0.2469726\ldots$ [1, §8], and we compute 0.24697261 for the same program, discretized as in [1].

3. The configuration

Write points as complex numbers. X_{30} consists of the 23 points of [1, Table 1], in their order, and the 7 points of Table 1. All coordinates lie in $\mathbb{Q}(\sqrt{3},\sqrt{11})$; we compute in this field exactly, with the basis $1, \sqrt{3}, \sqrt{11}, \sqrt{33}$. The unit-distance graph of X_{30} has 80 edges (that of X_{23} has 47) and 84 893 independent sets.

k	$\operatorname{Re} x_k$	$\operatorname{Im} x_k$	points at distance 1
24	$\frac{3-\sqrt{33}}{6}$	$\frac{\sqrt{3}}{3}$	3, 15, 20, 22, 25, 27
25	$\frac{11-\sqrt{33}}{12}$	$\frac{-\sqrt{3}+\sqrt{11}}{12}$	6, 15, 21, 22, 24, 30
26	$\frac{1}{6}$	$-\frac{\sqrt{11}}{6}$	2, 11, 12, 21
27	$\frac{9-\sqrt{33}}{6}$	$\frac{\sqrt{3}}{3}$	4, 7, 10, 16, 20, 24
28	$\frac{4}{3}$	$\frac{3\sqrt{3}+\sqrt{11}}{6}$	3, 5, 17, 18, 22
29	$\frac{2-\sqrt{33}}{6}$	$\frac{2\sqrt{3}+\sqrt{11}}{6}$	12, 13, 18, 19, 22
30	$\frac{17-\sqrt{33}}{12}$	$\frac{5\sqrt{3}+\sqrt{11}}{12}$	2, 6, 12, 25

Table 1. The points $x_{24}, \ldots, x_{30}$ added to X_{23}, and the points of X_{30} at distance 1 from each.

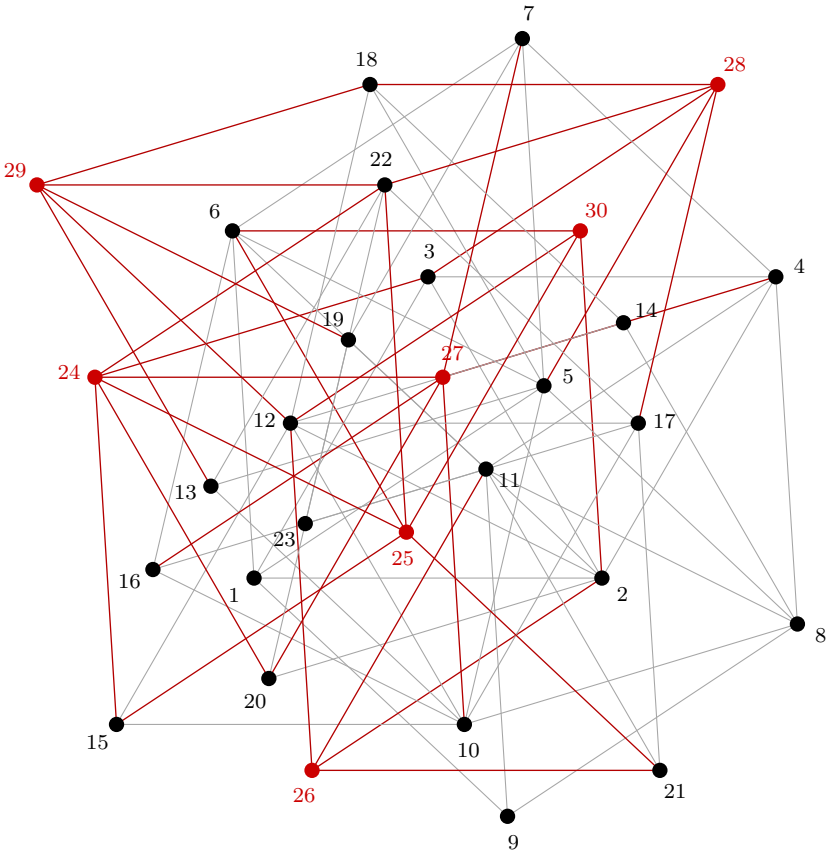


Figure 1. The configuration X_{30} and its 80 unit distances. The 23 points of [1] are black, the 7 new points red; red segments are the unit distances at a new point.

4. The certificate and its verification

The program for X_{30} has the inclusion–exclusion rows of [1]: one variable for each independent set, the equations $\Pr(i \in S) = \delta$ and $\Pr(i, j \in S) = f(r_{ij})$, and the congruence relations. We generate the latter ourselves: the independent sets with at least three points fall into congruence classes (compared exactly), and each class $S_1, \dots, S_m$ gives the $m - 1$ relations $\Pr(S_1 \subseteq S) = \Pr(S_j \subseteq S)$, with S_j listed in the order of a distance-preserving bijection. (Relations between pairs repeat the equations $\Pr(i, j \in S) = f(r_{ij})$.)

The certificate is found in two steps. First the program is solved in floating point, with κ supported on the grid $t \in \{0, 0.1, \dots, 100\} \cup \{100.25, 100.5, \dots, 400\}$; its multipliers w of the relations are rounded to multiples of 2^{-40} and frozen. For frozen w the best $\psi_0, \psi_T, \lambda, u$ solve a second, small program (one variable per point and per pair): minimize ψ_T subject to $Q \geq 0$ on every independent set, $-\sum_i \lambda_i \geq 1.03$, and $\Phi(t) \geq 1 + 2 \cdot 10^{-5}$ at the points of a grid of step 0.005 on $[0, 50]$ and 0.02 on $[50, 1000]$ where it is needed (rows are added at the local minima of Φ until none is violated). Then ψ_0, λ and u are rounded to multiples of 2^{-40} and ψ_T is raised until $Q \geq 0$ holds exactly on every independent set. The certificate for X_{30} uses 15 709 relations and gives $\psi_T < 0.24380$; it is the file `data/forbidden_distances/m1/certificates/m1_x30.json`. Nothing in this construction needs to be trusted: the checks below take the certificate as it is.

Two programs, written separately, check the certificate and prove Theorem 1:

- `check_m1.py` computes the unit-distance graph of X exactly in $\mathbb{Q}(\sqrt{3}, \sqrt{11})$, checks that every index lies in range, that no u_{ij} sits on a unit pair and that every relation in $\mathcal{C}$ is a congruence, checks $Q \geq 0$ on all independent sets with integers over a common denominator, and proves $\Phi \geq 1$ on $[0, T]$ by bisection with Taylor enclosures of J_0 in Arb ball arithmetic, and beyond T by $|J_0(x)| \leq \sqrt{2/(\pi x)}$; it cross-checks the number of independent sets by the recursion $I(G) = I(G - v) + I(G - N[v])$, and it does not rely on `assert` statements, which Python skips under its option `-O`.
- `check_m1_indep.py` has its own exact arithmetic, its own transcription-independent reading of the points, its own integer interval arithmetic with the power series and the Hankel expansion of J_0, and an adaptive sweep based on $|J_0''| \leq 1$; it computes Q on all independent sets by a subset-sum pass and recomputes a sample from the definition.

The facts about J_0 used are: $J_0(x) = \frac{1}{\pi} \int_0^\pi \cos(x \sin s)\, ds$, so $|J_0| \leq 1$ and $|J_0''| \leq 1$; $J_0 \geq -0.4028$, proved by `check_m1.py` on $[0, 4]$ by interval arithmetic and beyond by the next bound; and $|J_0(x)| \leq \sqrt{2/(\pi x)}$ for $x > 0$, which follows from Nicholson's formula $J_0(x)^2 + Y_0(x)^2 = \frac{8}{\pi^2} \int_0^\infty K_0(2x \sinh u)\, du$ [5, 10.9.30], [13, §§13.73–13.74].

Both accept the certificate (in under a minute) and the certificate for X_{23} alone, which gives 0.24698967 and reproduces the bound of [1]. Both reject corrupted copies: an index out of range (also in one-point relations), a broken congruence, a u_{ij} on a unit pair, a moved point, a lowered ψ_T, a raised λ_i, a lowered ψ_0. A referee report, written by a separate AI agent, proves Proposition 3 in full; its own checker, written before it read ours, accepts the four certificates, and of 28 corrupted certificates every invalid one is rejected by all three checkers. It found one weakness, now removed: `check_m1.py` relied on `assert` statements.

5. How the points were found

The search follows [1, §7]: a new point must lie at distance 1 from at least two points already present and at distance at least 0.1 from all of them. For X_{23} there are 208 such points. Adding one of them alone lowers the value of the program (in floating point, discretized with step 0.1 up to $t = 100$, where X_{23} gives 0.246948) to as little as 0.246083, and the gains of different points partly add up: the two best together gain $1.36 \cdot 10^{-3}$, four fifths of the sum of their single gains. The six best points at distance 1 from at least four points of X_{23}, added in the order of their single gains, give X_{29}, and x_{30} is the best of the others, with the third largest single gain (Table 2). Each point gains about $4 \cdot 10^{-4}$, slowly less as the set grows, while the program grows with the number of independent sets; we stopped at 30 points for the certificate.

points	independent sets	relations	value (floating point)	certified bound
23	13 552	2 122	0.246948	0.24698967
25	19 868	4 091	0.245583	–
26	28 622	5 008	0.245169	0.24522384
27	33 898	6 044	0.244841	–
29	55 610	10 012	0.244092	0.24414455
30	84 893	15 709	0.243743	0.24379682

TABLE 2. The program for X_{23} and its extensions: the value in floating point (with κ on the grid of Section 4), and the bound proved by a certificate (rounded up).

Remark. Ambrus et al. observed that the last point of their 24-point set did not improve the bound and removed it [1, §9]. Our points improve it from the first one on: the best single point lowers the value by $8.7 \cdot 10^{-4}$. We do not know how the announced bound 0.2415 [12] was obtained, so we cannot compare the methods.

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